

Problem Set 4
Winter 2020

Issued: Saturday, February 1

Due: Saturday, February 15

Full credit: 100 points.

Recall that a symmetric matrix $\Sigma \in \mathbb{R}^{d \times d}$ is

- positive semidefinite (PSD) if $a^T \Sigma a \geq 0$ for all $a \in \mathbb{R}^d$, and
- positive definite (PD) if $a^T \Sigma a > 0$ for all $a \in \mathbb{R}^d$ such that $a \neq 0$.

The two notions are denoted as $\Sigma \succeq 0$ and $\Sigma \succ 0$ respectively.

Problem 4.1 (15 points)

Show that the following distributions are exponential families: Exponential, Gamma, Beta, Poisson, and Geometric. In each case, specify the sufficient statistic(s), the natural parameter space, the log partition function and the dimension of the family.

Problem 4.2 (30 points)

Suppose that X has density in exponential family form

$$p(x; \theta) \propto \exp \left(\sum_{i=1}^d \theta_i T_i(x) - A(\theta) \right) h(x).$$

where each T_i is differentiable.

- (a) Prove the following form of *Stein's identity*. If X has support on the real line and g is any differentiable function such that $\mathbb{E}|g'(X)| < +\infty$, then

$$\mathbb{E} \left\{ \left[\frac{h'(X)}{h(X)} + \sum_{i=1}^d \theta_i T_i'(X) \right] g(X) \right\} = -\mathbb{E}[g'(X)].$$

- (b) Specializing to the $N(\mu, \sigma^2)$ case, use part (a) to show that $\text{cov}\{g(X), X\} = \sigma^2 \mathbb{E}[g'(X)]$.
- (c) Use part (b) to compute the third and fourth moments of the $N(\mu, \sigma^2)$ distribution.

Problem 4.3 (15 points)

Determine the natural parameter space of the associated exponential family of dimension one with $\mathcal{X} = \mathbb{R}$, $T(x) = x$, and

- (a) $h(x) = \exp(-x^2)$.
- (b) $h(x) = \exp(-|x|)$.
- (c) $h(x) = \exp(-|x|)/(1+x^2)$.

Problem 4.4 (15 points)

Consider an exponential family in canonical form with density

$$p_\theta(x) = h(x) \exp\{\langle \theta, T(x) \rangle - A(\theta)\}$$

Show that $\theta \mapsto A(\theta)$ is a convex function, by directly verifying the definition of convexity.

(Hint: You can use Hölder inequality:

The L^p space of functions ($p > 0$) with respect to ν is the set of functions f for which $\|f\|_p := [\int |f(x)|^p d\nu(x)]^{1/p}$ is finite. In compact notation $\|f\|_p := (\int |f|^p d\nu)^{1/p}$. The inequality says that if $f \in L_p$ and $g \in L_q$ with $1/p + 1/q = 1$ and $p, q > 0$, then

$$\|fg\|_1 \leq \|f\|_p \|g\|_q$$

This is a generalization of Cauchy–Schwarz which corresponds to the case $p = q = 2$. If you feel like proving Hölder, you can do it by using *Young inequality*: $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$ which holds for $a, b \geq 0$ (as long as $1/p + 1/q = 1$ and $p, q > 0$), and the monotonicity and linearity of integral/expectation.)

Problem 4.5 (5 points)

Prove Young inequality (see above) using the fact that $x \mapsto \log x$ is concave over $(0, \infty)$. A function f is concave if $-f$ is convex. This can be taken as the definition of concavity.

Problem 4.6 (15 points)

Consider the multivariate Gaussian distribution $N(\mu, \Sigma)$ with density

$$p(x) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right)$$

assuming that Σ is PD.

- (a) Show that $N(\mu, \Sigma)$ can be written as an exponential family in canonical form with sufficient statistics $\{X, XX^T\}$. That is, it can be written as

$$p_{\theta, \Theta}(x) \propto \exp\left\{\langle \theta, x \rangle + \langle \Theta, xx^T \rangle - A(\theta, \Theta)\right\}$$

where $\theta \in \mathbb{R}^d$, $\Theta \in \mathbb{R}^{d \times d}$ and $\langle \theta, x \rangle = \sum_i \theta_i x_i$ and

$$\langle \Theta, xx^T \rangle = \text{tr}(\Theta xx^T) = \text{tr}(x^T \Theta x) = x^T \Theta x.$$

Specify θ, Θ and $A(\theta, \Theta)$ in terms of the original parameters μ, Σ .

- (b) Is the matrix function $Z \mapsto -\log |Z|$ defined on the set of PD $d \times d$ matrices convex? Here, $|Z|$ is the determinant of matrix Z .
- (c) Is the function $z \mapsto z^T Q z$ defined on $\mathbb{R}^d$ convex when Q is PD matrix? (The answer in fact does not change if Q is PSD).

Problem 4.7 (20 points)

Let $f : \mathcal{X} \rightarrow \mathbb{R}$ be a real-valued function defined on some abstract space $\mathcal{X}$. Assume that $\mathcal{X}$ is linear, that is, for $x, y \in \mathcal{X}$ the operation $\alpha x + \beta y$ where $\alpha, \beta \in \mathbb{R}$, is well-defined. We say that f is convex, if the usual inequality is satisfied: $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$, for $x, y \in \mathcal{X}$ and $\lambda \in [0, 1]$.

- (a) For fixed $x, y \in \mathcal{X}$ let $\phi_{x,y} : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $\phi_{x,y}(t) = f(x + ty)$. Show that f is convex if and only if $\phi_{x,y}$ is convex for all $x, y \in \mathcal{X}$.
- (b) A differentiable function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is convex if and only if its derivative ϕ' is nondecreasing, that is, $(t - s)[\phi'(t) - \phi'(s)] \geq 0$ for all $t, s \in \mathbb{R}$. Show that a differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex if and only if $\langle x - y, \nabla f(x) - \nabla f(y) \rangle \geq 0$ for all $x, y \in \mathbb{R}^d$.
- (c) A twice differentiable function $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is convex if and only if its second derivative ϕ'' is nonnegative, that is, $\phi''(t) \geq 0$ for all $t \in \mathbb{R}$. Show that a twice differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex if and only if $\nabla^2 f(x) \succeq 0$ for all $x \in \mathbb{R}^d$, that is, its Hessian is PSD.
- (d) Show that a differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex if and only if $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$, for all $x, y \in \mathbb{R}^d$.

Comments: (b) and (d) are first-order characterizations of convexity. (b) shows that the map $x \mapsto \nabla f(x)$ is a monotone operator. (d) says that the convex function stays above its tangent plane at each point. (c) is a second-order characterization of convexity. A corresponding characterization of strict convexity is obtained if $\nabla^2 f(x) \succeq 0$ is replaced with $\nabla^2 f(x) \succ 0$.

Problem 4.8 (20 points)

Consider a random vector $X \in \mathbb{R}^d$. Let $\mu = \mathbb{E}[X] \in \mathbb{R}^d$ and

$$\Sigma := \text{cov}(X) = \mathbb{E}[(X - \mu)(X - \mu)^T] \in \mathbb{R}^{d \times d}$$

be the covariance matrix of X .

- (a) Show that $\Sigma = \mathbb{E}[XX^T] - \mu\mu^T$.

Show that Σ is positive semidefinite (PSD):

- (b) once by considering $\text{var}(a^T X)$ for some (nonrandom vector) $a \in \mathbb{R}^d$, and
- (c) once via the Jensen inequality. (Hint: Consider the function $f_a(X) = a^T(XX^T)a$ for a fixed $a \in \mathbb{R}^d$. Show that f_a is convex.)

Show that Σ is positive definite (PD) if and only if:

- (d) there is no $v \in \mathbb{R}^d$ and $c \in \mathbb{R}$ such that $v^T X = c$ almost surely (a.s.).
(Hint: You can use the fact that for a random variable Y , $\mathbb{E}Y^2 = 0$ implies $Y = 0$ a.s..)

Show that in a full-rank exponential family $\nabla^2 A$ is PD. (This implies that A is strictly convex.)